The self-consistency of the primordial metric with diagonal sources

The issue raised by the author on 2026-10-16, why the metric of 2026-10-14 is not self-consistent with diagonal sources, the simplest remedy (the warped metric), its field equations, its exact solutions and what it changes for the fields

dirac16complex project (with Claude Opus 5.5)

September 2026

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Abstract

A diagonal metric solves the Einstein (or Einstein-Lovelock) field equations $\sum_k \alpha_k E_{(k)}{}^{\mu}{}_{\nu} + \Lambda\,\delta^{\mu}_{\nu} = \kappa\,T^{\mu}{}_{\nu}$ only if every off-diagonal component of the left-hand side equals the corresponding component of the source. The author's notebook states the same as a warning: all off-diagonal source terms must vanish for the earlier solutions for Ψ_{16} to be valid. This document records what the Revision computation found. (1) The metric of 2026-10-14, with the second function $b_4(x_4)$ in the three extra-time components and $g_{44} = -1$, has at every Lovelock order the off-diagonal component $E_{(k)}{}^{x_4}{}_{x_0} \propto \cot z\, b_4'/b_4$; in Einstein gravity it is $\frac{3}{2} H \cot z\, b_4'/b_4$, so a diagonal source forces $b_4' = 0$. (2) In the two-parameter family $g_{00} = b_4^{2p} e^{2q a_4} \cot^2 z$, $g_{44} = -\sin^{2\gamma} z$ (every other entry unchanged) the Einstein component vanishes for all $a_4(x_4)$, $b_4(x_4)$ exactly when $q = 0$ and $6\gamma + 4p = 1$, and on that line the Gauss-Bonnet component is proportional to $(6\gamma{-}1)\, b_4'(4a_4'b_4 - b_4')^2$. The only member free of off-diagonal terms at every order is $\gamma = 1/6$, $p = 0$: the warped metric, which differs from the metric of 2026-10-14 in one entry, $g_{44} = -\sin^{1/3} z$, so that all seven directions $x_1, \ldots, x_7$ carry the same factor $\sin^{1/3} z$. (3) The warped metric is the warped product $ds^2 = dy^2 + e^{2Hy}\hat{g}_7$ of the hidden line ($y = \ln(\sin z)/(6H)$) and the 7-dimensional metric $\hat{g}_7$ of $x_1, \ldots, x_7$; every mixed Riemann component and every off-diagonal component of the Einstein, Gauss-Bonnet and cubic Lovelock tensors vanishes identically, for all a_4, b_4 and H (PROVED twice, with sympy and with Wolfram). (4) Its field equations split into layers $\sin^{-j/3} z$ times the Lovelock tensors of $\hat{g}_7$; Einstein gravity reduces exactly to the 7-dimensional Einstein equations on each slice of constant x_0, and with $\Lambda = -21H^2$ the Kasner metrics are exact vacuum solutions in which 3-space expands, the extra times deflate and $b_4 \propto x_4^{2/3}$ is not constant. With the author's weights there is no vacuum history with b_4 not constant; the record's history $a_4 = Hx_4$, $b_4 = b_4(0)e^{(4/3)Hx_4}$ needs a diagonal, conserved source, constant in time and layered in $\sin^{-1/3} z$. (5) For the fields the volume factor becomes $b_4^{3/2} \cos z \sin^{1/6} z$ and the gravity term of the Dirac operator $\frac{7}{2} H \gamma^{(x_0)} + \frac{3}{4}(b_4'/b_4) \sin^{-1/6} z\, \gamma^{(x_4)}$; a condensate that does not depend on x_0 exists only for $\lambda = 0$ and $m = \pm\frac{7}{2}H$. The off-diagonal components of the energy-momentum tensor of the fields themselves (the spin-connection bilinears) are a condition on the source that no choice of a diagonal metric removes; the Revision record recomputes them for the warped metric. The withdrawn alternative $g_{00} = b_4^{1/2} \cot^2 z$ makes only the Einstein tensor diagonal.

1 The issue

1.1 The author's statement

On 2026-10-16 the author wrote: "a diagonal form for the metric of the primordial gravitational field is only self-consistent if all off-diagonal terms of all source terms in the Einstein field equations ARE ZERO. This is not the case here." The author's notebook `WaveFunctionUniverse-4+4-Einstein-Lovelock.nb` (a scratch notebook of the author; it was read to understand the issue, and none of its definitions or conventions is used in the Revision record) carries the same warning next to the sources of the metric: the universe sources are to be added to the Einstein or Einstein-Lovelock field equations, and all their off-diagonal terms must vanish for the diagonal metric, and the earlier solutions for Ψ_{16}, to be valid. The notebook uses the metric without b_4 (the metric before 2026-10-14).

The order of 2026-10-16 asks for the simplest modification of the metric that is probably self-consistent with the Einstein field equations, for this document, and for the recomputation of every primordial object of the record.

1.2 Conventions of the Revision record

The coordinates are the author's $x_0, x_1, \ldots, x_7$: x_0 is the hidden direction, $z = 6Hx_0 \in (0, \pi/2)$, $H > 0$; x_1, x_2, x_3 are 3-space; x_4 is the time; x_5, x_6, x_7 are the three extra times, time-like and deflating. Every metric of this document is diagonal and depends on x_0 and x_4 only. The Riemann tensor is $R^a{}_{bcd}$ in the sign convention of Misner, Thorne and Wheeler, $R^{ab}{}_{cd} = g^{be} R^a{}_{ecd}$; the Lovelock tensors are

$$E_{(k)}{}^{h}{}_{j} = -\frac{1}{2^{k+1}}\, \delta^{hh_1\ldots h_{2k}}_{jj_1\ldots j_{2k}}\, R^{j_1 j_2}{}_{h_1 h_2} \cdots R^{j_{2k-1} j_{2k}}{}_{h_{2k-1} h_{2k}},$$

with the generalized Kronecker delta; $E_{(1)}$ is the Einstein tensor, $E_{(2)}$ the Gauss-Bonnet tensor, $E_{(3)}$ the cubic Lovelock tensor. The field equations are $\sum_{k=1}^{3} \alpha_k E_{(k)}{}^{\mu}{}_{\nu} + \Lambda\, \delta^{\mu}_{\nu} = \kappa\, T^{\mu}{}_{\nu}$ (`Revision /field_equations_a4/a4-equations.json`, conventions). The author's weights are $\alpha_k = w_k / H^{2k}$ with $w = (1, 1/32, 1/1152)$. We write $a_4' = da_4/dx_4$, $u = b_4'/b_4$.

2 Where off-diagonal terms come from

2.1 The geometry

For a diagonal metric $g = \mathrm{diag}(\eta_{\mu\mu} f_\mu^2)$ whose factors depend on x_0 and x_4 only, every Riemann component is of the form $R^{ab}{}_{ab}$ (a sectional curvature) except the mixed components $R^{x_0 c}{}_{x_4 c}$ and $R^{x_4 c}{}_{x_0 c}$, $c \notin \{x_0, x_4\}$. For factors that separate, $f_c = A_c(x_4) B_c(x_0)$, the mixed component is proportional to

$$\partial_0 \ln f_c \, (\partial_4 \ln f_c - \partial_4 \ln f_0) - \partial_4 \ln f_c \, \partial_0 \ln f_4.$$

The off-diagonal components of every Lovelock tensor are built from these mixed components, multiplied by sectional curvatures of the remaining directions; the only ones that can be nonzero are $E_{(k)}{}^{x_4}{}_{x_0}$ and $E_{(k)}{}^{x_0}{}_{x_4}$.

In the metric of 2026-10-14, $f_0 = \cot z$ and $f_4 = 1$ do not depend on the other variable, and the six factors $f_i = e^{a_4} \sin^{1/6} z$, $f_t = e^{-a_4} b_4^{1/2} \sin^{1/6} z$ all have $\partial_0 \ln f_c = H \cot z$. The mixed component of a 3-space direction is proportional to a_4', that of an extra time to $-a_4' + u/2$. The Einstein tensor adds the six, and the terms with a_4' cancel while those with u do not:

$$E_{(1)}{}^{x_4}{}_{x_0} = \frac{3}{2}\, H \cot z\, \frac{b_4'}{b_4}.$$

The Gauss-Bonnet and cubic components are $\cot z\, u$ times polynomials in a_4', u and H; with the weights α_k the sum is $\cot z\, u\, Q$ with the record's factor Q (PROVED: check `record_metric_of fdiagonal_reproduced`, reproducing `generalSource.offDiagonal_x4x0` of the record). A diagonal source therefore forces $u = 0$ in Einstein gravity, and with the author's weights on the history $a_4 = Hx_4$ it allows only $d \ln b_4 / da_4 \in \{0, 4/3\}$, where $Q = 0$.

2.2 The fields

The energy-momentum tensor of the fields has off-diagonal components of two kinds (the Revision record, `Revision/docs/DIRAC16COMPLEX00_FIELD_THEORY.md`, and the textbook, chapter 9):

components with one index x_0 and one index x_4, $T^{x_4}{}_{x_0} = -\frac{1}{4}(B_{40} - \cot z\, B_{04})$, built from derivatives along x_0 and x_4; they vanish for configurations that depend on x_4 only (check ***`commuting_T_x4x0_homogeneous`****);* components produced by the spin connection, which drops out of the Lagrangian but not of the tensor, for example $K^{x_4}{}_{x_1} = \frac{1}{2} e^{a_4} \sin^{1/6} z\, H\, \bar{\Phi}\gamma^{(4)}\gamma^{(1)}\gamma^{(0)}\Phi$ for a configuration that depends on x_4 only (check `spin_connection_in_the_energy_momentum_tensor`). For a generic condensate 42 of the 56 off-diagonal components are not zero; special condensates (the textbook's witnesses) make all of them vanish.

These components belong to the source. No diagonal metric removes them: a self-consistent diagonal metric needs a source whose off-diagonal components vanish, and a source that does not have this property needs a metric that is not diagonal in the corresponding directions.

2.3 What follows for the metric of 2026-10-14

Einstein gravity: a diagonal source forces $b_4' = 0$, which the user excluded ("b4 is never 1"). The author's weights: a diagonal source allows the single non-constant branch $d \ln b_4 / da_4 = 4/3$ on $a_4 = H x_4$, with the constant source $\kappa(\rho, p_3, p_t, p_0) = (-2/27, -2/9, -22/27, -14/27)$; no condensate can be that source (check `condensate_cannot_source_branch`).

The metric of 2026-10-14 is therefore self-consistent with diagonal sources only on one tuned branch of one theory. The remedy must remove the geometric off-diagonal term for every history and every Lovelock order; the field's own off-diagonal terms remain a condition on the source.

3 The simplest remedy: the warped metric

3.1 The family and the conditions

The candidates change one entry each, in the two places where the mixed component can be cancelled: the hidden direction, which may depend on x_4, and the time direction, which may depend on x_0. The family

$$g_{00} = b_4^{2p}\, e^{2q a_4} \cot^2 z, \qquad g_{44} = -\sin^{2\gamma} z$$

(every other entry as in the metric of 2026-10-14) contains the metric of 2026-10-14 ($p = q = \gamma = 0$). Two results, each PROVED with sympy and, independently, with Wolfram:

The only off-diagonal Einstein component is $E_{(1)}{}^{x_4}{}_{x_0} = -\frac{3H \cot z}{2b_4} \sin^{-2\gamma} z \left(4q\, a_4' b_4 + (6\gamma + 4p - 1)\, b_4'\right)$. It vanishes for all a_4, b_4 exactly when $q = 0$ and $6\gamma + 4p = 1$ (check ***`family_einstein_condition`****).* On that line ($q = 0$, $p = (1 - 6\gamma)/4$) the Gauss-Bonnet component is $E_{(2)}{}^{x_4}{}_{x_0} = \frac{3}{2} H \cot z\, \sin^{-4\gamma} z\, (6\gamma - 1)\, b_4' (4a_4' b_4 - b_4')^2 / b_4^3$. It vanishes for all a_4, b_4 exactly when $\gamma = 1/6$, $p = 0$ (check `family_gauss_bonnet_condition`).

So among these one-entry changes exactly one is free of off-diagonal terms at the Gauss-Bonnet order, and section 3.3 shows that it is free of them at every order:

$$ds^2 = \cot^2 z\, dx_0^2 + \sin^{1/3} z \left(e^{2a_4}(dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 \right.$$
$$\left. - e^{-2a_4}\, b_4 \,(dx_5^2 + dx_6^2 + dx_7^2) \right).$$

This is the **warped metric** of 2026-10-16. It differs from the metric of 2026-10-14 in one entry, $g_{44} = -1 \to -\sin^{1/3} z$: the time direction now carries the same factor $\sin^{1/3} z$ as the six other directions $x_1, x_2, x_3, x_5, x_6, x_7$. The functions $a_4(x_4)$ and $b_4(x_4) > 0$ are free; b_4 enters only through $u = b_4'/b_4$, so a constant factor of b_4 is a rescaling of x_5, x_6, x_7. The 3-space scale factor is $e^{a_4} \sin^{1/6} z$, the extra-time scale factor $e^{-a_4} b_4^{1/2} \sin^{1/6} z$, and the time lapse $\sin^{1/6} z$.

3.2 The warped-product form

With the proper coordinate of the hidden direction, $y = \ln(\sin z)/(6H) \in (-\infty, 0)$, one has $dy = \cot z\, dx_0$ and $\sin^{1/3} z = e^{2Hy}$, so (check `proper_hidden_coordinate`)

$$ds^2 = dy^2 + e^{2Hy}\, \hat{g}_7, \qquad \hat{g}_7 = e^{2a_4}(dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 - e^{-2a_4} b_4 (dx_5^2 + dx_6^2 + dx_7^2).$$

This is a warped product of the hidden line with the 7-dimensional metric $\hat{g}_7$, which depends on x_4 only. The coordinate y is the one the Kohn-Sham record already uses. The volume factor is $\sqrt{|g|} = b_4^{3/2} \cos z \, \sin^{1/6} z$ (check `sqrt_det_g`). The tip $z \to 0$ lies at $y \to -\infty$, at infinite proper distance, where the factor e^{2Hy} of all seven directions tends to 0.

3.3 Every off-diagonal component vanishes

For the warped metric, $\partial_0 \ln f_c = \partial_0 \ln f_4 = H \cot z$ for every $c \in \{x_1, \ldots, x_7\}$ and $\partial_4 \ln f_0 = 0$, so the expression of section 2.1 vanishes for every c: there is no mixed Riemann component at all (check `warped_mixed_riemann_vanish`: 28 nonzero components, all of the form $R^{ab}{}_{ab}$, against 12 mixed components in the metric of 2026-10-14). The off-diagonal components of every Lovelock tensor are built from mixed components, so all 56 off-diagonal components of $E_{(1)}$, $E_{(2)}$ and $E_{(3)}$ vanish identically, for all $a_4(x_4)$, $b_4(x_4)$ and H (check `warped_offdiagonal_all_orders`; Wolfram, the same check name). This is the general property of warped products over a line (B. O'Neill, *Semi-Riemannian Geometry*, chapter 7): the curvature has no mixed components between the base and the fiber.

3.4 The alternative that was not chosen

On the Einstein line the member $\gamma = 0$, $p = 1/4$ keeps $g_{44} = -1$ and changes the hidden direction instead, $g_{00} = b_4^{1/2} \cot^2 z$. Its Einstein tensor is diagonal, but $E_{(2)}{}^{x_4}{}_{x_0} = -\frac{3}{2} H \cot z\, b_4'(4a_4' b_4 - b_4')^2 / b_4^3$ and $E_{(3)}{}^{x_4}{}_{x_0}$ do not vanish (check `alternative_hidden_breathing_einstein_only`), and b_4 itself, not only u, enters the curvature through $H^2 b_4^{-1/2}$. It is self-consistent with diagonal sources only for Einstein gravity, so it is not used.

A metric with off-diagonal entries could absorb off-diagonal sources. The sources of section 2.2 have off-diagonal components in many directions (3-space with time, 3-space with the extra times, the hidden direction with the others); absorbing them would need off-diagonal metric entries in all these directions and would give up the homogeneity and isotropy of 3-space and of the extra times. That is not a simple modification, and it is not pursued.

4 The field equations of the warped metric

4.1 Layers

Write $W = \sin^{-1/3} z$ (the inverse square of the warp factor $\sin^{1/6} z$) and let $\hat{E}_{(j)}$ be the Lovelock tensor of order j of $\hat{g}_7$, computed separately in seven dimensions, with $\hat{E}_{(0)} = 1$; for the hidden direction let $\hat{E}^{(0)}_{(j)} = -2^{-(j+1)}\,\delta^{h_1\dots h_{2j}}_{j_1\dots j_{2j}}\,\hat{R}\cdots\hat{R}$ be its scalar. Then (check `warped_layer_decomposition`, sympy and Wolfram)

$$E_{(k)}{}^{\mu}{}_{\mu} = \sum_{j=0}^{k} c_{kj}\, H^{2(k-j)}\, W^{j}\, \hat{E}_{(j)}{}^{\mu}{}_{\mu},$$

with $c_{10} = 21$, $c_{11} = 1$; $c_{20} = -420$, $c_{21} = -40$, $c_{22} = 1$; $c_{30} = 2520$, $c_{31} = 360$, $c_{32} = -18$, $c_{33} = 1$. The constants c_{k0} are those of the maximally symmetric space of curvature $-H^2$ ($\hat{g}_7$ flat). The field equations become

$$\beta_0 + \sum_{j=1}^{3} \beta_j\, W^{j}\, \hat{E}_{(j)}{}^{\mu}{}_{\nu} = \kappa\, T^{\mu}{}_{\nu},$$

with $\beta_0 = 21\alpha_1 H^2 - 420\alpha_2 H^4 + 2520\alpha_3 H^6 + \Lambda$, $\beta_1 = \alpha_1 - 40\alpha_2 H^2 + 360\alpha_3 H^4$, $\beta_2 = \alpha_2 - 18\alpha_3 H^2$, $\beta_3 = \alpha_3$. The author's weights give $\beta_0 = 161/16 + \Lambda$ (in the units of the weights), $\beta_1 = 1/(16H^2)$, $\beta_2 = 1/(64H^4)$, $\beta_3 = 1/(1152H^6)$ (check `warped_field_equations_layers`).

4.2 Einstein gravity reduces to seven dimensions

For Einstein gravity ($\alpha_1 = 1$, $\alpha_2 = \alpha_3 = 0$) the field equations read $(21H^2 + \Lambda)\,\delta^{\mu}_{\nu} + W\,\hat{G}^{\mu}{}_{\nu} = \kappa\, T^{\mu}{}_{\nu}$, where $\hat{G}$ is the Einstein tensor of $\hat{g}_7$ for $\mu, \nu \in \{x_1, \dots, x_7\}$ and $\hat{G}^{x_0}{}_{x_0} = -\hat{R}/2$. With $\Lambda = -21H^2$ every self-consistent source has the form $T^{\mu}{}_{\nu} = \sin^{-1/3} z\; \tau^{\mu}{}_{\nu}(x_4)$, and the equations are exactly the 7-dimensional Einstein equations $\hat{G}^{\mu}{}_{\nu} = \kappa\,\tau^{\mu}{}_{\nu}$ on every slice of constant x_0, together with the hidden pressure $\kappa\,\tau^{x_0}{}_{x_0} = -\hat{R}/2$. The sectional curvatures are

$$\hat{G}^{x_4}{}_{x_4} = \frac{3}{4}(4a_4'^2 - 2a_4' u - u^2), \qquad \hat{G}^{x_1}{}_{x_1} = -\frac{1}{2}\Big(6a_4'^2 - 6a_4' u - 2a_4'' + 3\frac{b_4''}{b_4}\Big),$$

$$\hat{G}^{x_5}{}_{x_5} = -\frac{1}{4}\Big(12a_4'^2 + 4a_4'' + 4\frac{b_4''}{b_4} - u^2\Big), \qquad -\frac{\hat{R}}{2} = -\frac{3}{2}\Big(2a_4'^2 - a_4' u + \frac{b_4''}{b_4}\Big)$$

(the keys `fiber_lovelock_tensors` and `fiber_lovelock_scalars` of the results file `warped-metric.json`).

4.3 Exact vacuum solutions with b_4 not constant

For Einstein gravity with $\Lambda = -21H^2$ the vacuum equations are $\hat{G} = 0$: $\hat{g}_7$ must be Ricci-flat. The Kasner metrics are Ricci-flat; for $\hat{g}_7$ they are

$$e^{a_4} = x_4^{p_s}, \qquad b_4 = x_4^{2/3}, \qquad p_s = \frac{1+\sqrt{5}}{6},$$

with the extra-time scale factor $e^{-a_4} b_4^{1/2} = x_4^{(1-\sqrt{5})/6}$ (and the mirror solution with $p_s = (1-\sqrt{5})/6$). All 64 components of the field equations hold at every x_0 (check `einstein_vacuum_kasner`, sympy and Wolfram). In this solution 3-space expands as $x_4^{0.539}$ and the three extra times deflate as $x_4^{-0.206}$; in terms of the e-folds of 3-space the extra-time scale factor is $e^{-0.382\,a_4}$ and $d\ln b_4/da_4 = \sqrt{5} - 1$. It is an exact, self-consistent primordial solution of the Einstein equations with b_4 not constant and a vanishing source. Its expansion is a power of x_4, not an exponential.

With the author's weights the vacuum equations must hold at every x_0, so each layer vanishes separately: $\beta_0 = 0$ and $\hat{E}_{(1)} = \hat{E}_{(2)} = \hat{E}_{(3)} = 0$. The Kasner metrics have $\hat{E}_{(2)} \neq 0$, and an exponential $\hat{g}_7$ ($a_4 = AHx_4$, $b_4 = e^{\beta AHx_4}$) is never Ricci-flat ($\hat{E}^{(0)}_{(1)} = -\frac{3}{2}A^2H^2(\beta^2 - \beta + 2) \neq 0$). The warped metric therefore has no vacuum history of the author's theory with b_4 not constant (check `author_weights_vacuum_conditions`). The author's earlier vacuum ($b_4 = 1$, $a_4 = Hx_4$, a solution of the metric before 2026-10-14) is not a solution of the warped metric: its layers with W do not vanish (check `earlier_vacuum_not_warped_solution`).

4.4 The record's history and its source

The Revision record uses the prescribed history $a_4 = Hx_4$, $b_4 = b_4(0)\,e^{(4/3)Hx_4}$, on which the extra times deflate exponentially, as $e^{-a_4/3}$. In the warped metric this history is no longer singled out by an off-diagonal condition (every history has a diagonal left-hand side); the record keeps it for continuity, as a prescribed background. Its source is diagonal, isotropic in 3-space and in the extra times, constant in time, and layered in $W = \sin^{-1/3} z$ (check `history_source_einstein`; the values are `history_source` of `warped-metric.json`, reproduced by Wolfram in `history_source_equals_sympy`):

left-hand side $\sum_k \alpha_k E_{(k)}{}^{\mu}{}_{\mu}$	**Einstein gravity**	**the author's weights (units of the weights)**
$\mu = x_4$: $-\kappa\rho - \Lambda$	$21H^2 - \frac{1}{3}H^2W$	$\frac{161}{16} - \frac{W}{48} + \frac{W^2}{48} + \frac{5W^3}{432}$
$\mu = x_1, x_2, x_3$: $\kappa p_3 - \Lambda$	$21H^2 - \frac{5}{3}H^2W$	$\frac{161}{16} - \frac{5W}{48} + \frac{17W^2}{432} + \frac{W^3}{432}$
$\mu = x_5, x_6, x_7$: $\kappa p_t - \Lambda$	$21H^2 - \frac{13}{3}H^2W$	$\frac{161}{16} - \frac{13W}{48} + \frac{11W^2}{144} - \frac{7W^3}{432}$
$\mu = x_0$: $\kappa p_0 - \Lambda$	$21H^2 - \frac{11}{3}H^2W$	$\frac{161}{16} - \frac{11W}{48} + \frac{53W^2}{432} - \frac{13W^3}{432}$

So for Einstein gravity with $\Lambda = -21H^2$ the history needs $\kappa\rho = \frac{1}{3}H^2 \sin^{-1/3} z > 0$, $\kappa p_3 = -\frac{5}{3}H^2 \sin^{-1/3} z$, $\kappa p_t = -\frac{13}{3}H^2 \sin^{-1/3} z$ and $\kappa p_0 = -\frac{11}{3}H^2 \sin^{-1/3} z$. For a diagonal tensor whose

entries depend on x_0 and x_4 the warped metric gives $\nabla_\mu T^\mu{}_\nu = 0$ identically for $\nu \neq x_0, x_4$, and both remaining components vanish for these sources, as the Bianchi identity requires (check `history_source_conserved`). The history remains a PRESCRIBED background: no field of the record is shown to be its source.

5 What the warped metric changes for the fields

5.1 The vielbein, the volume and the gravity term

The diagonal vielbein is $f = (\cot z,\ e^{a_4}\sin^{1/6} z\ (\times 3),\ \sin^{1/6} z,\ e^{-a_4} b_4^{1/2}\sin^{1/6} z\ (\times 3))$, the volume factor $b_4^{3/2}\cos z\,\sin^{1/6} z$. The gravity term of the Dirac operator, $\gamma^\mu\Omega_\mu = \frac{1}{2}\sum_a \gamma^{(a)}\frac{1}{\sqrt{|g|}}\partial_a(\sqrt{|g|}/f_a)$, is

$$\gamma^\mu\Omega_\mu = \frac{7}{2}\,H\,\gamma^{(x_0)} + \frac{3}{4}\,\frac{b_4'}{b_4}\,\sin^{-1/6} z\,\gamma^{(x_4)}$$

(check `dirac_gravity_term`, sympy and Wolfram); in the metric of 2026-10-14 it was $3H\gamma^{(x_0)} + \frac{3}{4}(b_4'/b_4)\gamma^{(x_4)}$. The hidden coefficient counts seven warped directions instead of six. The anticommutation $\{\gamma^\mu, \Omega_\mu\} = 0$ for each direction separately, which the record proves for every diagonal vielbein, still removes the spin connection from the Lagrangian.

5.2 Condensates

The field equation of a configuration $\Phi(x_4)$ that does not depend on the other coordinates reads, in the warped metric,

$$\sin^{-1/6} z\,\gamma^{(x_4)}\Big(\partial_4 + \frac{3}{4}\frac{b_4'}{b_4}\Big)\Phi + \frac{7}{2}H\gamma^{(x_0)}\Phi = V\Phi, \qquad V = m + \lambda S.$$

Multiplying by $\sin^{1/6} z$, the left side $\gamma^{(x_4)}(\partial_4 + \frac{3}{4}u)\Phi$ does not depend on x_0, and the right side $\sin^{1/6} z\,(V - \frac{7}{2}H\gamma^{(x_0)})\Phi$ is $\sin^{1/6} z$ times a column that does not depend on x_0. Both must vanish: $\Phi = b_4^{-3/4}\Phi_0$ with a constant column Φ_0, and $\gamma^{(x_0)}\Phi_0 = (2V/7H)\Phi_0$. Because $(\gamma^{(x_0)})^2 = 1$, $V = \pm\frac{7}{2}H$ is constant, and since $S = b_4^{-3/2}\Phi_0^\dagger C\Phi_0$ changes with b_4, this needs $\lambda = 0$ (or $S = 0$). A condensate that does not depend on x_0 therefore exists only for $\lambda S = 0$ and $m = \pm\frac{7}{2}H$, with $\gamma^{(x_0)}\Phi_0 = \pm\Phi_0$ (derived in this section; the record re-derives the solutions of the field equations for the warped metric). In the metric of 2026-10-14 condensates existed for every m and λ; in the warped metric the fields depend on the hidden direction, through the profiles that the warped product separates.

5.3 The field's own off-diagonal components

The spin-connection bilinears of section 2.2 are present in the warped metric as well (for example $K^{x_4}{}_{x_1}$ contains $H\,\bar\Phi\gamma^{(4)}\gamma^{(1)}\gamma^{(0)}\Phi$). A self-consistent source must have vanishing off-diagonal components; this is a condition on the configurations of the fields (or on the quantum states whose

expectation values are the source), which the recomputation of the field-theory record for the warped metric states component by component. At this commit that condition is OPEN for the warped metric.

6 Consequences for the Revision record

The warped metric replaces the metric of 2026-10-14 in `Revision/SPEC.md` section 1 (the user's order of 2026-10-16: "Recompute all primordial objects"). Everything that depends on the metric is recomputed with it: the field theory of both fields (spin connection, field equation, energy-momentum tensor, conservation, exact solutions, quantisation), the field equations for a_4 and b_4 (Lovelock tensors, sources, the history's source), the lead's checks, the generalized-Kronecker-delta record, the dark sector, the Kohn-Sham record (the hidden-direction mode problem now carries the lapse $\sin^{1/6} z$), the pairing theorems, the notebooks, the documents and the textbook. The metric of 2026-10-14 and its results remain in the repository as the withdrawn earlier record, quoted only for comparison.

7 What is proved, computed, assumed and open

7.1 Proved (exact symbolic computation, twice: sympy and Wolfram)

The metric of 2026-10-14 has the off-diagonal pair (x_4, x_0) at every Lovelock order, equal to the record's $\cot z\, u\, Q$*.* In the family of section 3.1 the Einstein component vanishes for all a_4, b_4 iff $q = 0$, $6\gamma + 4p = 1$, and the Gauss-Bonnet component on that line iff $\gamma = 1/6$, $p = 0$. *The warped metric has no mixed Riemann component, and every off-diagonal component of $E_{(1)}$, $E_{(2)}$, $E_{(3)}$ vanishes identically.* The layer decomposition, the effective couplings, the reduction of Einstein gravity to seven dimensions, the Kasner vacuum solutions with $\Lambda = -21H^2$, the absence of a vacuum history of the author's theory with b_4 not constant, the history's source and its conservation, the volume factor and the gravity term of the Dirac operator.

7.2 Computed

Nothing numerical: every statement of this document is an exact identity.

7.3 Assumed

The history $a_4 = Hx_4$, $b_4 = b_4(0)e^{(4/3)Hx_4}$ is a prescribed background, kept for continuity with the record; its source is computed, not derived from a field. "Simplest" is understood as: one entry of the metric changed, within the family of section 3.1; other families (several entries, or a metric that is not diagonal) are not exhausted.

7.4 Open

Which configurations of dirac16complex00, and which quantum states of dirac16complex, have an energy-momentum tensor without off-diagonal components in the warped metric, and whether any of them can be the source of a history (section 5.3). Whether a source of the layered form of section 4.4 arises from the fields. * A history of the author's theory that is driven by the fields themselves.

8 Reproduction

Run from the repository root (each run stays below 60 s):

```
python Revision/metric_warped/python/derive_warped_metric.py
wolframscript -file Revision/metric_warped/wolfram/verify_warped_metric.wls
python scripts/build_provenance_pdf.py Revision/docs/METRIC_SELF_CONSISTENCY.md \
    --developer-layout --specifications Revision/pdf-specifications.json
python -m unittest Revision/tests/test_metric_self_consistency_publication.py -v
```

command	checks	time
derive_warped_metric.py	16 of 16 PASS	45 s
verify_warped_me tric.wls	11 of 11 PASS	44 s
build_provenance_pdf.py	the PDF build	6 s

The last lines of the two programs begin with `python_warped_metric=OK` and `wolfram_warped_m etric=OK`.

Two runs of each program give byte-identical outputs. The provenance of the record, with the sha256 of every input and output, is `Revision/metric_warped/WARPED_METRIC_PROVENANCE.md`.

9 Verification records

report	checks
Revision/metric_warped/reports/p ython-warped-metric-report.json	16 of 16 PASS
Revision/metric_warped/reports/w olfram-warped-metric-report.json	11 of 11 PASS

check	sympy	Wolfram
sqrt_det_g	PASS	PASS
proper_hidden_co ordinate	PASS	
record_metric_of fdiagonal_reproduced	PASS	PASS

check	sympy	Wolfram
family_einstein_condition	PASS	PASS
family_gauss_bonnet_condition	PASS	PASS
warped_mixed_riemann_vanish	PASS	PASS
warped_offdiagonal_all_orders	PASS	PASS
alternative_hidden_breathing_einstein_only	PASS	PASS
warped_layer_decomposition	PASS	PASS
warped_field_equations_layers	PASS	
einstein_vacuum_kasner	PASS	PASS
author_weights_vacuum_conditions	PASS	
history_source_einstein	PASS	
history_source_equals_sympy		PASS
history_source_conserved	PASS	
dirac_gravity_term	PASS	PASS
earlier_vacuum_not_warped_solution	PASS	